

Unit 1 — Introduction to AI & Ethics, and Core Ethical Principles

Strong & weak AI • Descriptive / Normative / Meta ethics • Ethics & law • Machine ethics • Moral diversity & testing • Bias & fairness, transparency & explainability, privacy & security, robustness & reliability

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Source: Bartneck, Lütge, Wagner, Welsh — *An Introduction to Ethics in Robotics and AI* (Springer, 2021), Ch. 2–4. Items tagged

✚ are short additions beyond the textbook for syllabus topics the book treats lightly.

1. What is AI?

Who	Definition
Kaplan & Haenlein	A system's ability to correctly interpret external data, learn from it, and use that learning to achieve goals through flexible adaptation.
Poole & Mackworth	Study of computational agents that act intelligently : actions fit circumstances & goals, flexible, learn from experience, choose well under limits.
Russell & Norvig	Study of agents that receive percepts and take actions — each agent is a function mapping percepts → actions.

- **Four schools** (Russell & Norvig): machines that **think like humans, act like humans, think rationally, act rationally**.
- Common core: designing **intelligent agents that achieve goals**. AI is a *moving target* — "AI is whatever computers can't do yet".
- **Autonomy** is context-dependent (Greek *auto-nomos*, "self-rule"): engineering → works without an operator; bioethics → patient's right to decide; Kantian ethics → choosing one's own moral rules.

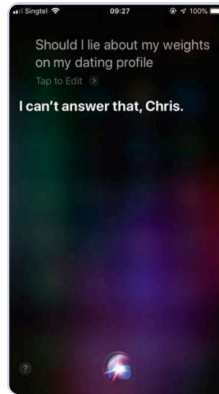


Figure 2.1 — Siri: "I can't answer that, Chris." (Should I lie about my weight on my dating profile?)

Why this "simple" question is hard

Needs speech recognition → language understanding (homophone *lie*) → world knowledge (dating, honesty) → a **moral stance**: **consequentialist** (maximise everyone's happiness), **deontological** (lying violates duty, Kant), **virtue** (be honest, Aristotle) — plus business risk for Apple.

1.1 Machine learning

Type	Idea	Example
Supervised	Learn from labelled data (classification/regression) → produces a <i>classifier</i> . Evaluated by true-positive and false-positive rate on a separate test set .	Spam filter
Unsupervised	Find structure without labels (clustering, PCA); exploratory.	Customer segments
Reinforcement	Learn a policy (action → expected reward) from a reward signal.	Robot navigating to a goal

Ethics link

Supervised labels are assigned by humans, so **societal bias can be baked into the classifier** — the thread that leads into Unit 2.

1.2 Robots & why AI is hard

- **Robot** = AI that is **situated** (real world) and **embodied** (physical body). Works by **Sense** → **Plan** → **Act**; an error at one step propagates to the next.
- **System integration** (sensors/actuators with different failure modes) is one of the hardest parts.
- **Hard for AI**: limited sensing; **poor generalisation** (face recogniser fails on a profile view); **no common sense / common ground**; no innate knowledge (a robot is a "blank slate"); repeated **hype cycles**.
- **Science vs fiction**: HAL, Terminator, Westworld inflate expectations (the "Frankenstein complex"). **Sophia** (Saudi "citizenship", 2017) is scripted PR — no learning, no adaptation.

2. Strong vs Weak AI & Types of AI Systems

Turing Test (1950)

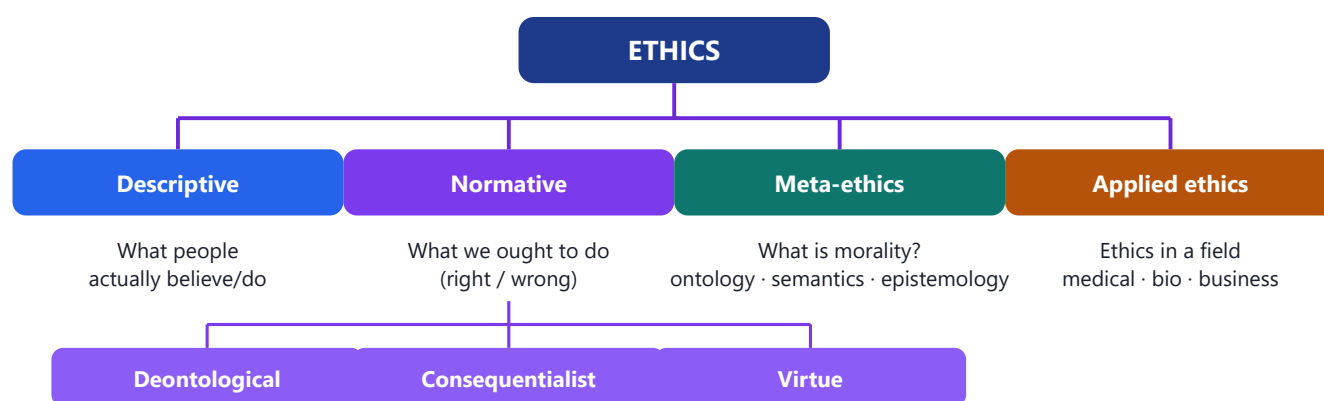
If a human interviewer cannot tell a machine from a person in conversation, the machine is judged intelligent. Replaces the vague word "thinking" with a testable task. *Criticism*: no time limit specified; passing is not clearly necessary/sufficient for intelligence.

	Weak (narrow) AI	Strong AI (Searle, 1980)
Scope	One narrowly defined task	General intelligence across many problems
Claim	Simulates intelligent behaviour	A suitably programmed computer <i>has a mind</i> like humans
Status	All current AI (chess/Go engines, loan scoring, Siri)	Not yet achieved

Types of AI systems: **Expert systems** (rule-based; e.g. mortgage advisors) • **Planning systems** (e.g. SPIKE for the Hubble telescope) • **Computer vision** (object recognition) • **Machine learning**.

3. Types of Ethics

Ethics = theory of morality (principles, general norms); **morality** = the actual rules, values and norms guiding action. Kant: ethics asks "*What should I do?*" (Greek *ethos* = custom; Cicero's *mores* → "moral").



Taxonomy of ethics (Ch. 3)

Branch	Question	Key points
Descriptive	What do people actually believe and do?	Empirical — moral psychology, experimental economics. E.g. the ultimatum game : people give up profit for fairness. Gives input to normative ethics ("experimental ethics").
Normative	What <i>ought</i> to be done?	Prescriptive; claims general validity ("stealing is wrong for everybody"). Three theories below.
Meta-ethics	What is morality itself?	Ontology (what has moral worth), semantics (meaning of right/wrong/ought), epistemology (how moral truths are known). "Theory of ethics".
Applied	Ethics in concrete fields	Medicine, biotech, business. Influence runs both ways : practice shapes theory (solidarity means sharing in a family, fair competition in society).

3.1 The three normative theories

Theory	Judges an action by	Detail	CSR example
Deontological (duty ethics)	The action itself — intention, duty, rule	Kant's categorical imperative : act only on a maxim you can will as a universal law without contradiction.	Critics ask about the company's motive (PR vs genuine)

Consequentialist	Foreseeable consequences	Bentham & Mill's utilitarianism — maximise happiness.	Motive irrelevant; only the impact on society counts
Virtue ethics	The agent's character	Plato's cardinal virtues (wisdom, justice, fortitude, temperance); Aristotle extends to 11 moral + intellectual virtues.	Is the firm acting honestly and justly?

4. Relationship Between Ethics and Law

Popular view: "*ethics starts where the law ends*" — the book challenges it:

1. **Laws have an ethical side** — anti-pollution and anti-trust laws are also ethical norms.
2. **Ethics works as "soft law"** — firms follow ethical standards the law doesn't demand, to protect reputation and share price (e.g. *fair-trade* coffee as a selling point); the effect is almost like **hard law**.

Remember

Law = enforceable minimum; ethics = broader, often precedes and shapes law (e.g. hate-speech laws on social media). They overlap but are not opposites.

5. Machine Ethics

Machine ethics

Asks: **what would it take to build an ethical AI that makes moral decisions?** Machines lack **phenomenology** (feelings, consciousness) and moral intuition — they only process *data about* feelings. Critics (van Wynsberghe & Robbins) say roboticists haven't given strong reasons to build moral robots.

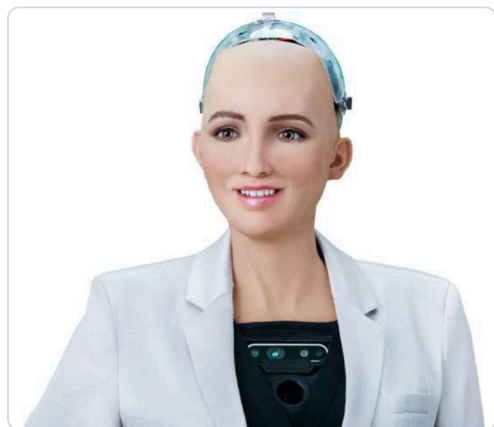


Figure 3.3 — Sophia: lifelike, but no feelings or consciousness

Sensors → Symbol grounding → Moral cognition (logic) → Action

- **Symbol grounding** = converting raw sensor data into symbols (objects, events, properties, relations).
- **Logic** = truth-preserving inference (Socrates is a man; all men are mortal ⇒ Socrates is mortal).

5.1 Machine ethics examples

Easy case — speeding-ticket robot

Rule: *if driver X is speeding then the robot is obligated to issue a ticket to X*. Fact: X is speeding ⇒ issue ticket. Clear rule + clear symbols = easy.

Hard case — toddler vs letter

A robot on its way to post a letter sees a toddler fall into a 1 m stream. Two duties clash: **rescue** vs **post**. The robot has no empathy or sense of urgency, so it needs **causal understanding** and a **utility scale**: toddler = +1,000,000, on-time letter = +1 $\Rightarrow$ rescue.

When naive utilitarian arithmetic fails

Now the robot drives a truck with **1,000,001 letters** (+1 each) $\Rightarrow 1,000,001 > 1,000,000 \Rightarrow$ it lets the toddler drown. Simple utility sums give **counter-intuitive** results; implementing deontology needs a principled way to **resolve clashes of duties**.

6. Moral Diversity and Testing

- **Core problem: no agreement on the correct moral theory** — polling (Bourget & Chalmers 2014): roughly $\frac{1}{4}$ of philosophers favour deontology, $\frac{1}{4}$ consequentialism, $\frac{1}{2}$ virtue ethics. Which one do we implement?
- **Approach — test cases**: design many ethical test cases; moral competence = ability to pass them. Iterating on new cases may expand competence and give insight into theory.
- **Testing & certifying fairness (active research)**: IEEE standards project on **algorithmic bias** (2017); **AI Fairness 360, audit-AI**; services like O'Neil Risk Consulting & Algorithmic Auditing; Facebook's **Fairness Flow**.
- Also: codes of ethics for robotics engineers and HRI professionals. Some researchers remain **pessimistic** about machine morality.

7. Trust — the Foundation of the Core Principles

Trust (Lee & See 2004)

The attitude that an agent will help achieve one's goals **in a situation characterised by uncertainty and vulnerability**.

- Trust drives **user acceptance**: untrusted products fail. **Culture** matters — e.g. after the US-EU *Safe Harbor* was struck down (2015), cloud firms opened EU data centres to regain trust.
- **Functional elements**: reliability, dependability, false-alarm rate, transparency, task complexity — **performance & reliability dominate**. Risk of injury/loss of life makes people reconsider trust.
- **Ethical elements**: the five AI4People principles below. With AI it is unclear whether the machine decides or follows a script, so human experience of trust may mislead.

8. The Five AI4People Principles (Floridi et al. 2018)

Non-maleficence

do no harm

Beneficence

do good

Autonomy

respect wishes

Justice

be fair, unbiased

Explicability

be explainable

Principle	Meaning & examples
Non-maleficence	AI shall not harm people, animals or property. Online bullying (Sweden's 1993 law) and hate speech (Germany 2017: fines up to €50 M if illegal content isn't removed within a week — but risks over-removal vs free speech).

Beneficence	AI shall do good ; benefits must outweigh harms (from bioethics): fewer AV accidents, elder-care robots, telemedicine, smart grids, protecting biodiversity, personalised education.
Autonomy	AI shall respect people's goals and wishes . Allow informed risk-taking (Everest sherpas, rock climbing) but not harm to others (child tells a robot "pull my sister's hair"); be wary of euthanasia robots. Kant's three formulations : universal law • humanity as an end, never merely a means • kingdom of ends. Asimov's Three Laws alone don't make a Kantian ethical agent.
Justice	AI shall act justly and without bias → Section 9.
Explicability	It must be possible to explain why an AI reached a result → Section 10.

9. Bias and Fairness (Principle of Justice)



Figure 4.1 — Justitia: blindfold = impartiality, scales = weighing evidence, sword = punishment

- Defining "justice" for a machine is hard because **moral theory is contested**; but in a **narrow scope with clear laws**, AI can make specific moral decisions.
- Credit scoring**: neighbourhood data → systematic bias against residential areas.
- Healthcare algorithm (Obermeyer 2019)**: race excluded, yet **cost** used as proxy for **need** → poorer patients under-served; root cause = **problem formulation**.
- COMPAS** (courts): ProPublica (2016) found bias against African-American defendants; developers cited differing **base rates**. Chouldechova (2017): predictive parity and equal error rates **cannot both hold** when base rates differ → **fairness definitions conflict**, and there is a **fairness – accuracy trade-off**. Accuracy only ≈65%.

Detail in Unit 2

Types of bias, detection metrics and mitigation techniques are covered in the Unit 2 notes.

10. Transparency and Explainability (Principle of Explicability)

- Explicability = intelligibility + accountability** (Floridi). It is **not identical to transparency**: publishing millions of lines of code won't help non-experts and may harm competitiveness.
- Intelligibility**: no inscrutable **black box** — someone can explain the workings to judges, juries and users.
- Legal**: EU **GDPR "right to information/explanation"** for algorithmic decisions — hard for neural networks → research in **Explainable AI (XAI)**.
- Justification** is part of moral functioning. *Loomis v. Wisconsin*: defendant challenged the proprietary COMPAS score; appeal failed as judges did not rely on the score alone.
- Accountability: log files** (like an aircraft flight recorder, "black box") let investigators retrace steps and assign blame.

11. Privacy and Security +

Key ideas	
Privacy	Control over personal data. AI is data-hungry; book Ch. 8 covers persistent surveillance, use of data for unintended purposes, auto-insurance discrimination, China's Social Credit System. Principles: consent, purpose limitation, data minimisation , anonymisation, GDPR rights.

Security

Protect data and models from breaches, **adversarial attacks**, **data poisoning** and model theft. Failures destroy trust and cause harm (links to non-maleficence).

12. Robustness and Reliability +

- **Robustness:** correct behaviour under changed, noisy or adversarial inputs — the book stresses AI **generalises poorly** and lacks common sense.
- **Reliability:** consistent, dependable performance over time; the book names it (with performance) the **dominant factor in trust**. Failures in safety-critical domains (vehicles, medicine) risk life.
- **Practices:** stress and edge-case testing, drift monitoring, fail-safe fallbacks, **human oversight**.

13. Quick Revision & Likely Questions

One-liners

- AI = agents that perceive, learn and act towards goals; today's AI is all **weak**.
- Ethics: **descriptive** (is) • **normative** (ought) • **meta** (nature) • **applied** (fields).
- Normative: **deontology** (duty, Kant) • **consequentialism** (outcomes, utility) • **virtue** (character, Aristotle).
- Ethics ≠ law but overlap; ethics acts as **soft law**.
- Machine ethics hurdles: no feelings, **no agreed theory**, naive utilities misfire → **test cases**.
- Five principles: **Non-maleficence, Beneficence, Autonomy, Justice, Explicability**.

Likely questions

1. Differentiate strong and weak AI with examples. (2)
2. Explain descriptive, normative and meta-ethics. (6)
3. Compare deontological, consequentialist and virtue ethics with one example. (8)
4. Discuss the relationship between ethics and law. (5)
5. Using the toddler–letter example, explain difficulties in machine ethics. (6)
6. How does moral diversity affect testing of ethical AI? Name bias-testing tools. (5)
7. Explain the five AI4People principles; relate justice and explicability to COMPAS. (10)